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Report on

An Emotion-Aware Digital Twin for Human-Centric Urban Infrastructure Planning

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DECLARATION

We, the undersigned students, hereby declare that the project titled “An Emotion-Aware Digital Twin for Human-Centric Urban Infrastructure Planning” submitted in partial fulfillment of the requirements for PROJECT BASED LEARNING (PBL-2) LAB (CSP297) is our original work.

This project has been carried out by us as a team during the B.Tech 2nd Year, Semester 4 (Session 2025–2026) under the guidance of our faculty. We confirm that this work has not been submitted previously, either in part or in full, to any other institution or for any other academic purpose.

We further declare that each member has actively contributed to the development of this project, and we take full responsibility for the authenticity, accuracy, and originality of the work presented.

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Project Title:

An Emotion-Aware Digital Twin for Human-Centric Urban Infrastructure Planning

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ABSTRACT

Use of digital twin technology in urban settings is becoming more prevalent in relation to infrastructure planning, environmental analysis, and smart city development. Current digital twin technology is mainly concerned with physical and functional aspects of infrastructure, such as traffic flow, pollution, and energy efficiency, and the human experience is still an unexplored area. The urban experience is not only about infrastructure but also about comfort, stress, and feelings of safety.

In this paper, a perception-aware digital twin framework for human-centric urban infrastructure planning is proposed. The framework consists of a system that models an urban area using a map-based digital twin and environmental factors such as traffic density, noise level, temperature, and green cover. A perception modeling part of the framework computes an Urban Perception Index (UPI) based on weighted environmental factors and machine learning algorithms. A scenario simulation part of the framework enables the assessment of infrastructure plans by monitoring their impact on perceived comfort and stress.

The proposed approach extends traditional digital twin systems by incorporating human perception into simulation and planning processes, enabling more socially informed urban infrastructure decisions.

1. INTRODUCTION

Rapid urbanization and smart city development have increased the need for intelligent tools capable of modeling complex urban environments. Digital twin technology has been identified as an efficient method for modeling virtual replicas of physical infrastructure systems, allowing simulation, monitoring, and decision-making in urban planning applications [1]. Urban digital twins are applied in transportation optimization, environmental monitoring, and infrastructure performance assessment [2].

However, despite these developments, the majority of digital twin applications are still centered on technical efficiency and performance. Cities, as experienced by their inhabitants, have a set of perceptual and psychological aspects, in addition to their physical structures. Environmental factors such as noise levels, crowd density, temperature, and green areas have a substantial impact on perceived comfort and stress. Studies on smart city governance have emphasized the need to incorporate emotional and perceptual information into the city system, although this information is not typically used in digital twin models [3].

Digital twins have also been used in sustainability planning and green infrastructure monitoring, proving their worth in supporting policy decisions and infrastructure performance tracking [4]. However, these systems currently lack the ability to assess human emotional response to infrastructure changes. This creates a gap between infrastructure optimization and human-centric urban planning.

To fill this research gap, this study proposes a perception-aware digital twin model that combines environmental factors and artificial intelligence methods to simulate perceived levels of comfort and stress in urban settings. The proposed model allows for scenario simulation of infrastructure projects and the generation of perception heat maps to inform planning decisions. Through the integration of perception models with digital twin models, this proposed research contributes to the development of smart cities that are livable, sustainable, and human-centric [5].

2. LITERATURE SURVEY

2.1 Overview of Digital Twin Technology in Urban Systems

Digital twin technology allows for the development of virtual replicas of physical systems for monitoring, simulation, and decision-making. Batty [1] explains digital twins as dynamic models of cities that combine infrastructure data, environmental data, and simulation engines. Fuller et al. [6] further clarify digital twins as systems that combine real-world data with computational models for prediction and optimization. Rosen et al. [8] and Grieves and Vickers [12] set the conceptual framework for digital twins as tools for managing complex systems through continuous data integration and simulation. The above studies show the potential of digital twin technology in the planning of urban infrastructure.

2.2 Digital Twins in Smart City Applications

Digital twins are being increasingly applied in the context of smart cities for infrastructure management, environmental studies, and urban analytics. Liu et al. [3] have proposed an urban digital twin framework that combines geospatial data with simulation models for planning purposes. Biljecki et al. [6] have described the application of geospatial modeling approaches for the development of urban digital twins. Zhang et al. [10] have reviewed digital twin technologies for smart cities and emphasized their role in sustainability analysis and transportation optimization.

2.3 Digital Twins in Simulation and Infrastructure Planning

Another strength of digital twin systems is the simulation ability. Wang et al. [4] illustrate the use of geospatial technologies in simulating infrastructure scenarios and analyzing planning decisions. Kumar et al. [11] examine the use of digital twins in sustainable urban planning and green infrastructure performance analysis. These studies illustrate the ability of digital twins to allow planners to analyze infrastructure decisions virtually before implementation.

2.4 Human-Centric Smart Cities and Urban Perception

Recent studies have underlined the significance of human-centric strategies in the development of smart cities. Anthopoulos [2] has pointed out the significance of taking into account the experience of citizens in urban planning systems. Smith et al. [5] have suggested an emotion-data framework for incorporating emotional factors into smart city governance. Nasar [5] has shown the significance of perceived safety and environmental factors in human comfort in urban areas.

2.5 Environmental Factors Affecting Urban Perception

Factors such as temperature, availability of green spaces, noise levels, and density of urban areas have a great impact on human comfort. The World Health Organization states that green spaces in urban areas have a positive effect on mental and physical health [9]. Li et al. [3] discuss the effect of urban heat islands on environmental comfort. Research on AI-based urban analytics indicates that environmental factors can be computationally modeled to analyze urban conditions [4].

2.6 Artificial Intelligence in Urban Analytics

Artificial intelligence methods are being increasingly applied for the analysis of urban settings and decision-making. AI-based urban analytics solutions can analyze environmental and infrastructure data to forecast trends and assess urban conditions [4]. Such computational models form the basis for perception modeling in digital twin settings, where environmental inputs can be converted into perception indicators.

2.7 Digital Twins for Human-Centric Urban Planning

Digital twins offer a platform that can be used to integrate simulation of infrastructure with environmental analysis. Batty [1] and Liu et al. [3] explain how digital twins can be used to support planning decisions through simulation of scenarios. Kumar et al. [11] illustrate how digital twins can be used to improve the evaluation of sustainability in urban infrastructure projects. However, most digital twin systems that have been developed are based on technical performance indicators and not human perception.

2.8 Challenges in Perception-Aware Digital Twin Systems

- Lack of perception integration in digital twins: The majority of current digital twin solutions are centered on infrastructure performance metrics such as traffic, energy, and pollution, while human perception metrics such as comfort and stress are not considered.
- Difficulty in quantifying human perception: Perception factors like comfort, safety, and stress are difficult to quantify using computational models because they involve emotional and psychological aspects.
- Data availability and consistency: Availability of data like noise level, crowd density, and availability of green spaces may not be consistent in all regions.
- Integration of heterogeneous data sources: Integration of geospatial data, environmental data, and perception models into a single digital twin framework may pose technical difficulties.
- Limited real-time perception datasets: Most research work related to perception is based on surveys or theoretical models, not real-time environmental data.
- Scalability of digital twin systems: Scaling up digital twin models from small urban areas to large city-scale environments is a challenge.
- Validation of perception models: Validation of the predicted levels of comfort and stress to accurately represent human experience is a challenge in perception-enabled simulation systems.

2.9 Summary of Literature Insights

The literature survey review indicates that the use of digital twin technology is prevalent in the simulation of cities, monitoring of infrastructure, and smart city analysis. Artificial intelligence has enhanced the prediction of the environment and decision-making in urban systems. Studies on human-centric smart cities indicate the significance of perception and well-being in the environment. The use of perception modeling in digital twin technology is still limited. This serves as a motivation for the development of a perception-aware digital twin system that integrates environmental information, artificial intelligence modeling, and scenario simulation for human-centric urban infrastructure planning [1]–[6], [10], [11].

3. PROBLEM STATEMENT

Urban planning systems and digital twin technologies are being utilized for simulating infrastructure performance, environmental monitoring, and smart city decision-making processes. The existing digital twin models are mostly focused on physical and operational parameters such as traffic flow, pollution levels, energy consumption, and infrastructure efficiency. However, these smart city technologies rarely include human perception parameters such as comfort, stress, safety, and environmental satisfaction.

The urban experience is not only related to the performance of urban infrastructure but is also associated with various environmental and psychological factors, such as noise levels, crowd density, temperature conditions, lighting quality, and green spaces. These factors have a major impact on the quality of life. However, most of the digital twin platforms being developed today do not take these human factors into account.

Thus, urban planning decisions that are optimal in technical terms can still be harmful to human well-being and livability in cities. The lack of perception-aware evaluation tools prevents urban planners from effectively evaluating the impact of changes in city infrastructure on citizens before implementation.

Therefore, there is a need to design a system in which indicators related to environmental perception are incorporated in digital twin models. The aim of this research is to design a Perception-Aware Digital Twin system in which models related to environmental factors are created, human comfort and stress levels are predicted, and infrastructure scenarios are simulated.

4. SYSTEM REQUIREMENTS

4.1 Functional Requirements

- The system should collect and process environmental parameters such as noise levels, traffic density, temperature, crowd density, lighting conditions, and green space availability.
- It should create a digital twin representation of the selected urban environment and calculate an Urban Perception Index (UPI) to estimate human comfort and stress levels.
- The system should allow simulation of different infrastructure scenarios and generate visual outputs such as perception scores and heatmaps for analysis.

4.2 Non-Functional Requirements

- The system should provide accurate and reliable perception analysis with efficient data processing.
- It should offer a user-friendly interface suitable for planners and researchers.
- The system should be scalable so that additional environmental parameters or simulation scenarios can be integrated in the future.

4.3 Software Requirements

- Programming Language: Python
- Libraries and Tools: Pandas, NumPy, Scikit-learn, Plotly/Matplotlib, Folium
- Development Environment: VS Code / Jupyter Notebook
- Dashboard Framework: Streamlit

5. METHODOLOGY

The proposed system utilizes a structured methodology to model an urban environment using a perception-aware digital twin system. This methodology includes data collection from the environment, data processing, perception indicator computation, and simulation of infrastructure scenarios to assess their effects on human comfort and stress levels.

5.1 Urban Environment Modeling

The first step is to define the boundary of the chosen urban area and then create a virtual version of the urban area. The urban environment consists of zones or grid sections, where the environmental parameters include the level of noise, temperature, density of the traffic, density of the crowd, quality of light, and the availability of green space. This virtual version of the urban environment is called the digital twin model.

5.2 Environmental Data Processing

The collected environmental data is preprocessed to ensure consistency and accuracy. Data cleaning and normalization techniques are applied to standardize the parameters. Min-Max normalization is used to scale the environmental parameters to a uniform range, ensuring that all parameters contribute proportionally to the perception modeling process.

5.3 Perception Index Calculation

After this preprocessing, the system uses an Urban Perception Index (UPI) calculation to determine the comfort and stress levels of humans in the environment. The system uses a weighted perception model in which various environmental factors like noise, traffic, temperature, and green spaces are assigned weights according to their influence on human perception. Linear regression methods are also used for this purpose.

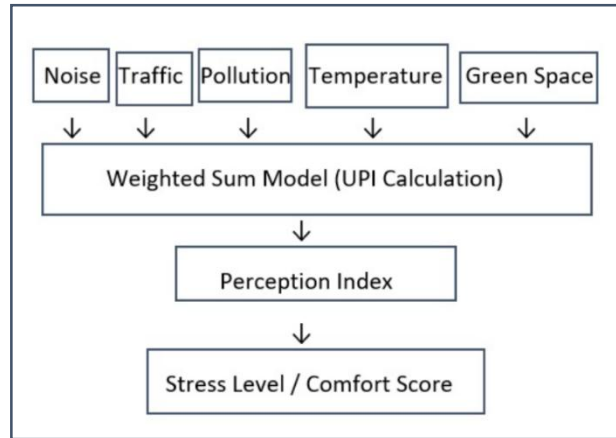


Fig: 5.1 Perception Model Structure

The diagram illustrates how external conditions like noise, vehicular traffic, pollution, heat, and greenspace are integrated through the use of a weighted sum method to compute the Urban Perception Index (UPI). The UPI is subsequently employed to ascertain the total amount of stress or degree of comfort experienced in the urban setting.

5.4 Scenario Simulation

The system can simulate various scenarios for the infrastructure to test potential decisions in urban planning. The system utilizes a rule-based method for simulation in altering environmental parameters during the simulation of scenarios. For instance, more green space may reduce stress levels, while more traffic density may increase perceived stress levels. The system will automatically compute the perception scores after each scenario.

5.5 Visualization and Decision Evaluation

The final stage is the visualization of the perception results in the form of dashboards and heatmaps. These visualizations will help identify areas where stress levels are high or comfort levels are low in the digital twin model. These results can be used to assess the viability of any proposed infrastructure and choose the best options to improve livability in the city.

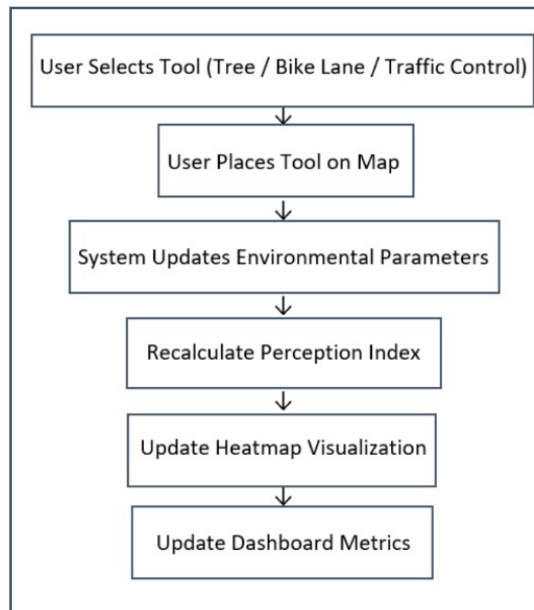


Fig 3.2 User Interaction and System Response Flow

The user interaction flow shows how the system dynamically reacts to the actions of the user within the digital twin environment. For example, when the user applies an intervention, such as planting trees or reducing traffic, the corresponding parameters of the environment are updated and the perception index is recalculated.

6. SYSTEM DESIGN AND IMPLEMENTATION

The proposed perception-aware digital twin system aims at integrating environmental information, perception modeling, and scenario simulation under a unified framework. The proposed system consists of different interconnected modules that can handle environmental parameters, simulate infrastructure changes, and display perception outcomes.

6.1 System Architecture

The proposed system is a perception-aware digital twin that will be used to simulate environmental conditions prevailing in an urban setting, which will eventually affect human comfort as well as stress. The proposed system will integrate various environmental conditions such as pollution, traffic density, temperature, and availability of green spaces to create a visual representation of urban stress.

The platform provides users with an opportunity to interact with a virtual city environment and make improvements to it, such as planting trees or controlling traffic. They can then see changes in stress distribution in real-time. This helps in evaluating different urban interventions

6.2 Workflow of the System

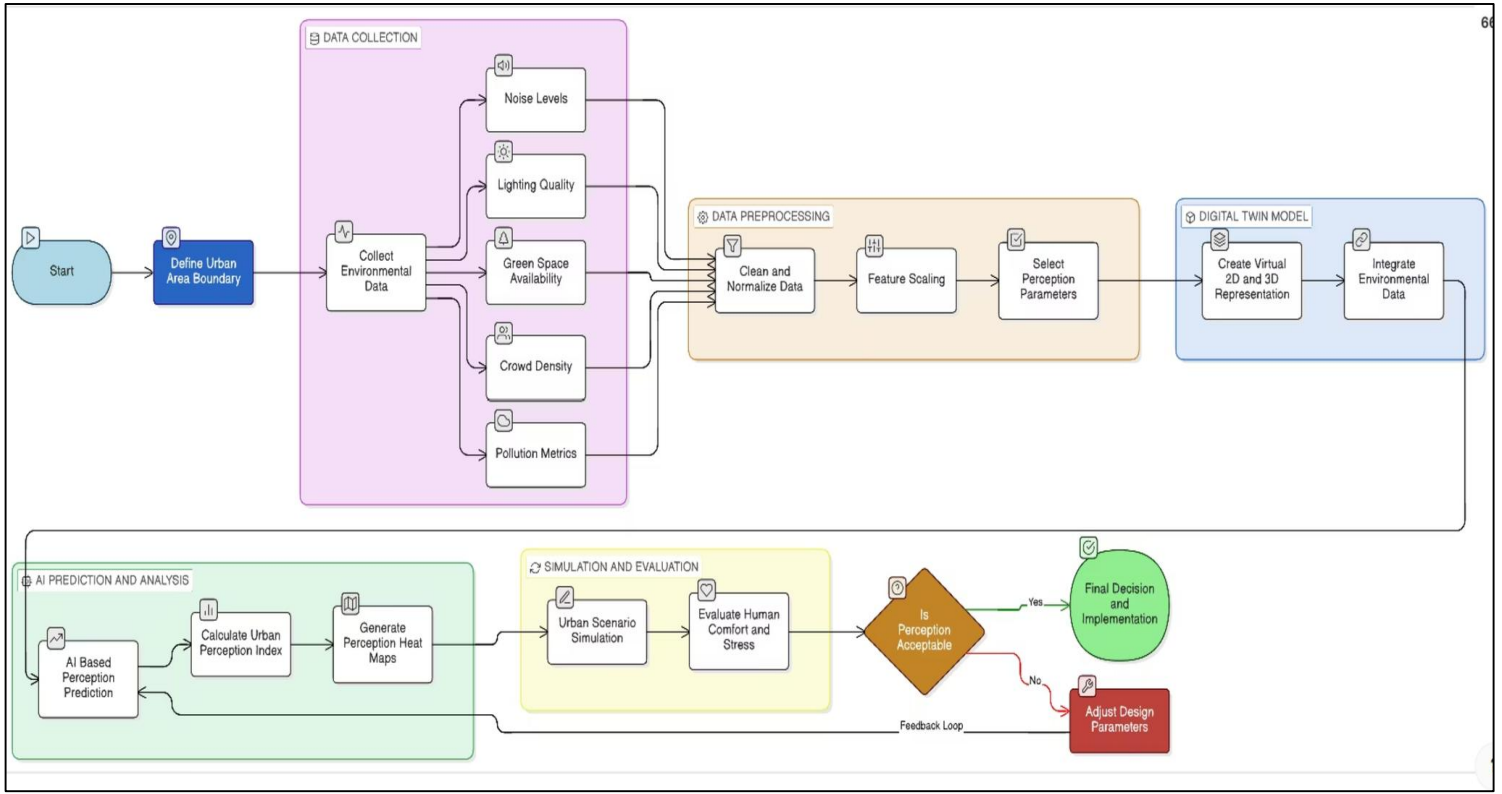


Fig 6.2.1: Workflow of the Perception-Aware Digital Twin System

Figure 6.2.1 shows the workflow of the proposed perception-aware digital twin system in human-centric urban infrastructure planning. The proposed workflow starts by specifying the scope of the selected urban area and collecting environmental information relevant to urban perception.

During the Data Collection stage, parameters such as the level of noise, quality of light, availability of green space, density of crowds, and levels of pollution are collected. These parameters define the most important environmental parameters that directly impact the level of comfort and stress of humans within an urban environment.

The collected data then goes to the Data Preprocessing stage, where the data is cleaned and normalized to maintain consistency in the data across all the parameters. Techniques like feature scaling are used to scale the features so that the variables can be compared and processed appropriately despite the scale or units in which they are measured. Parameters related to perception are then selected.

Next, the processed data is incorporated into the Digital Twin Model. This model creates a virtual version of the chosen urban environment. This version stores the attributes of the environment related to different zones. This model acts as a simulation tool for urban environment evaluation.

At the stage of AI Prediction and Analysis, perception modeling techniques are used to estimate the levels of human comfort and stress. The Urban Perception Index (UPI) is calculated, and perception heatmaps are generated to visualize the spatial variations in urban perception.

Next, the process will continue to the Simulation and Evaluation phase, in which different scenarios can be simulated for the infrastructure. Urban interventions, such as increasing green spaces, improving traffic flow, or lighting conditions, can be simulated in the digital twin environment. The system will evaluate the perception scores to check if the urban environment is comfortable enough.

Finally, a decision evaluation step is used to evaluate if the perception outcome is acceptable. If the perception outcome meets the required threshold, the system will proceed to the final decision and implementation stage. However, if the perception outcome does not meet the required threshold, the design parameters are adjusted, and the simulation process is repeated in a feedback loop.

This workflow enables planners to analyze how environmental changes influence human perception in urban environments and supports more human-centric urban infrastructure planning.

6.3 Algorithms Used

Module	Algorithm / Method	Why This Algorithm is Needed	Short Description
Data Preprocessing	Min–Max Normalization	Environmental inputs have different ranges	Scales values between 0 and 1 for fair comparison
Perception Modeling	Linear Regression	To predict stress/comfort from environmental factors	Models relationship between inputs (noise, traffic, etc.) and perception score
Perception Index Calculation	Weighted Sum Model	To combine multiple environmental factors	Calculates a single perception index using factor weights
Scenario Simulation	Rule-Based Simulation	To test infrastructure changes easily	Updates environmental values when scenarios change
Decision Evaluation	Threshold Decision Logic	To determine acceptable comfort level	Compares perception score with predefined limit

Fig 6.3.1: Relevant Algorithms

- **Min–Max Normalization:** Used to scale environmental variables into a consistent range during preprocessing.
- **Linear Regression:** Used to model the relationship between environmental factors and perception indicators.
- **Weighted Perception Index Model:** Combines environmental parameters into a single perception score.
- **Rule-Based Simulation Algorithm:** Modifies environmental parameters during scenario testing.
- **Threshold Decision Logic:** Determines whether perception levels fall within acceptable limits.

6.4 Implementation Tools

The system is implemented with Python as the programming language. Libraries such as Pandas and NumPy are used for data processing, and Scikit-learn is used for perception modeling. Plotly or Matplotlib libraries are used for generating charts and heatmaps. Streamlit is used for creating an interactive dashboard for simulating and visualizing the system in real time.

6.5 Prototype/ User Interface Design

The developed prototype offers an interactive and visually intuitive interface to simulate urban environments. The interface has three major components: the digital twin map, the tools panel, and the analytics dashboard.

6.5.1 Base Heatmap Visualization

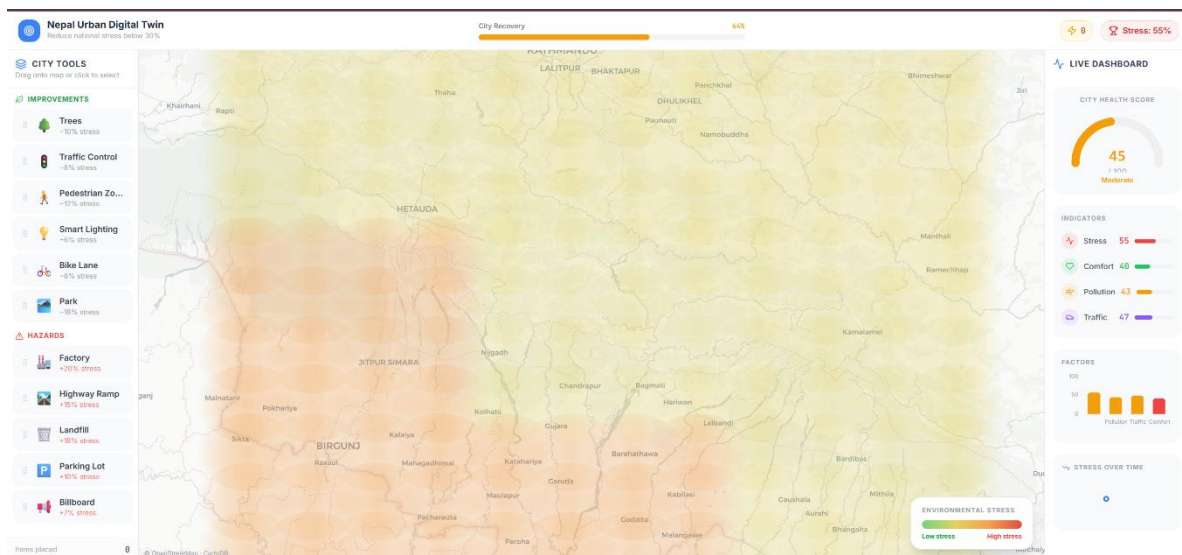


Fig 6.5.1: Base Heatmap Representation of Urban Stress Distribution

The above image represents the basic digital twin of the city using a heatmap overlay. The visualization represents the distribution of stress levels across different areas of the city without requiring further user interaction. The heatmap is represented using a range of colors as follows:

- The green color represents areas of low stress levels.
- The yellow color represents areas of moderate stress.
- The red color represents areas of high stress.

The semi-transparent overlay ensures that the underlying map features such as roads and geographical regions remain visible. This provides a realistic and interpretable representation of urban condition

6.5.2 Digital Twin Map with Interaction

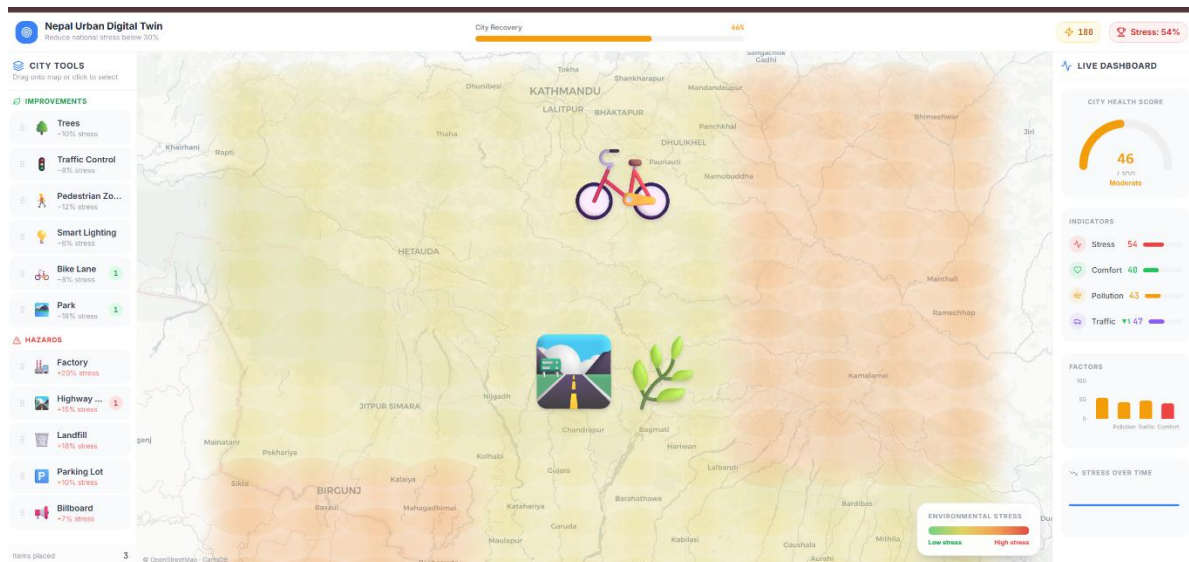


Fig 6.5.2: Interactive Digital Twin Map with Applied Urban Interventions

This image shows the interactive capability of the system. Users can add different urban improvement tools to the map. These include bike lanes, roads, and green spaces.

These elements dynamically affect the environmental conditions in these regions. The heatmap also updates to reflect the changes in stress levels. This allows users to carry out a scenario analysis. They can easily visualize the effects of their decisions.

6.5.3 Interactive Tools Panel

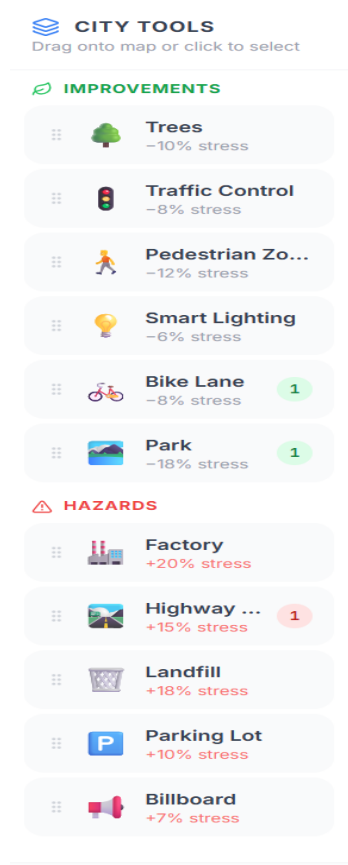


Fig 6.5.3: Urban Planning Tools and Hazard Controls Panel

The tools available on the left-hand panel are used to change the environmental conditions. The tools are grouped under two categories:

- Improvements: Trees, pedestrian zones, smart lighting, bike lanes, and parks, which are used to reduce stress levels
- Hazards: Factories, highways, landfills, parking, which are used to increase stress levels

Each tool has a specified impact value which allows for a realistic simulation.

6.5.4 Live Dashboard and Analytics

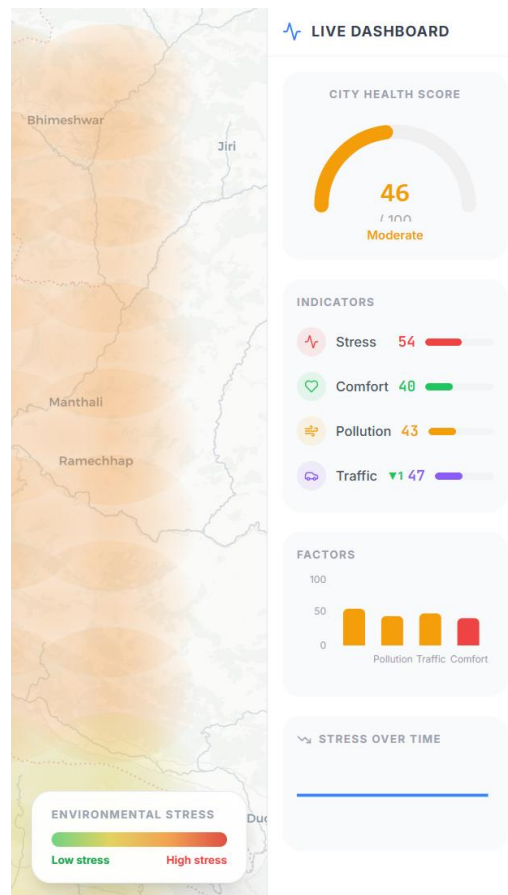


Fig 6.5: Real-Time Dashboard Showing City Health and Environmental Indicators

The right-side panel displays real-time analytics of the urban environment. It includes:

- City health score represented using a gauge meter
- Stress, comfort, pollution, and traffic indicators
- Graphical representation of environmental factors
- Stress trend over time

This dashboard provides a comprehensive overview of urban conditions and supports data-driven decision-making.

6.5.5 Interactive Simulation Behavior

The system provides a dynamic interactive behavior in which the actions taken by the user affect the environmental conditions. If improvements, like planting trees or adding bike lanes, are applied, the regions affected will demonstrate a gradual decrease in the level of stress.

This can be observed through smooth changes in colors in the heatmap. The system can be used as a tool for “what-if” analysis.

6.5.6 Design Features and Implementation Highlights

- The design features of the system include:
- Map-based digital twin representation of the environment
- Semi-transparent heatmap for representation of the stress
- Interactive tools for modification of the environment
- Real-time analytics dashboard
- Scenario-based simulation for evaluation of planning

All these design features provide a useful system for human-centric infrastructure planning.

7. RESULT AND DISCUSSIONS

7.1 Perception Visualization and Spatial Stress Distribution Analysis

The developed perception-aware digital twin prototype was successful in generating a visual representation of urban stress by using a heatmap-based digital twin map. The basic idea of the heatmap is to show the level of environmental stress present in the selected urban area. The different shades of color on the map represent the different levels of environmental stress present in the area. The regions with low levels of environmental stress are represented by the color green, the regions with moderate levels of environmental stress are represented by the color yellow, and the regions with high levels of environmental stress are represented by the color red.

From the prototype results generated by the developed digital twin prototype, it can be clearly seen that the digital twin is successful in generating a visual representation of environmental conditions and human perception in a simplified manner. The semi-transparent overlay of the heatmap allows users to get a clear view of geographical features like roads, districts, and regions. Hence, it can be clearly seen that the system is successful in converting environmental parameters into a spatial perception map for easy urban stress analysis.

7.2 Impact of Urban Improvements on Stress Reduction

There are several improvement tools in the prototype that include trees, bike lanes, parks, pedestrian areas, and smart lighting. Once these tools are placed on the digital twin map, the stress level in the environment is reduced accordingly.

For example, when bike lanes, roads, and green elements were placed on the digital twin map in the simulation process, the health score of the city increased from 45 to 46, and the stress level in the environment decreased from 55% to 54%. This indicates that the system dynamically updates the environmental stress level and perception scores accordingly.

Based on the results obtained in this simulation process, the findings are as follows:

- Trees and park areas significantly reduce stress in the environment.
- Bike lanes improve traffic flow.
- Smart lighting improves safety and comfort.
- Pedestrian areas reduce stress in traffic areas.

This indicates that the prototype has been successful in simulating real-world scenarios in the environment and has been beneficial in making appropriate decisions regarding infrastructure improvements.

7.3 Hazard Simulation and Stress Increase

The prototype also includes the addition of hazard elements such as factories, highways, landfills, parking lots, and billboards that cause increased environmental stress. Once these elements are incorporated in the map, the system increases the stress levels and reduces the city health score.

The results obtained from the system are as follows:

- Factories cause increased levels of pollution and stress
- Highways cause increased traffic density and noise
- Landfills cause increased environmental discomfort
- Parking lots cause increased traffic congestion
- Billboards cause increased visual pollution

This proves that the system can be used to simulate negative urban interventions and evaluate the impact on human comfort. The simulation of negative urban interventions can help city planners evaluate the potential risks involved in the construction process.

7.4 Live Dashboard and City Health Monitoring

The live dashboard, as shown on the right side of the prototype, is useful for monitoring real-time city health and environmental factors. The live dashboard includes the following information:

- City Health Score (46 - Moderate)
- Stress Level (54)
- Comfort Level (40)
- Pollution Level (43)
- Traffic Level (47)
- Environmental factor charts
- Stress over time graph

The results indicate that the live dashboard is able to update real-time information depending on the simulation actions performed. For instance, when improving factors are added, the stress and comfort levels are affected as shown in the live dashboard. When hazards are added, the stress and city health are affected.

The live dashboard is useful for monitoring urban conditions and making decisions regarding infrastructure instantly. The live dashboard offers a comprehensive overview of the urban environment.

7.5 Scenario Simulation and What-If Analysis

The prototype also allows for scenario-based simulation, where users can test different configurations of the infrastructure and observe their impact on stress and comfort levels.

For example:

- Adding green spaces reduces stress and enhances comfort
- Enhancing traffic control reduces pollution and other traffic-related factors
- Making a combination of improvements is better than making a single improvement, etc.

From the above, it is clear that integrated urban planning strategies are more effective in achieving better outcomes compared to isolated strategies. This is because, with this system, urban planners can carry out a "what-if" scenario and compare various scenarios to arrive at a better solution. Hence, this feature makes a powerful tool for urban planning and simulation.

7.6 Decision Support and Planning Insights

The results obtained from the prototype show that the system can help urban planners in decision-making by identifying high stress points and possible improvements.

The system has the following features:

- Visual stress distribution
- Environmental real-time indicators
- Impact of improvements and hazards
- Evaluation of city health
- Scenario comparison

The results obtained show that the digital twin with perception awareness can help in urban planning by considering environmental information and human perception in infrastructure decision-making.

The digital twin prototype has been successful in showing the potential of digital twin technology in improving urban livability and sustainability by considering human comfort and stress.

7.7 System Performance and Observations

The prototype was efficient in handling environmental parameters and providing simulation output in real-time. The system was able to update heatmaps, indicators, and dashboard data with minimal lag. The key observations from the implementation are as follows:

- The system is responsive to user actions
- Stress levels are dynamic and change according to environmental updates
- There are several improvements to obtain better output
- Green infrastructure has the maximum positive impact
- Traffic and pollution cause significant stress
- Visualization using heatmaps helps in better understanding of urban conditions

The results have validated the fact that perception-aware digital twin is capable of providing simulation output for urban planning.

8. CONCLUSION AND FUTURE WORK

CONCLUSION:

- The aim of this work was the proposal of a Perception-Aware Digital Twin for human-centric urban infrastructure planning.
- The proposed system incorporates environmental factors such as noise, traffic, temperature, and green space for the evaluation of comfort and stress levels.
- The Urban Perception Index (UPI) enables the measurement of human perception within an urban environment.
- Simulation scenarios were used for the evaluation of the impact of changes in the infrastructure on the perception results.
- Tools for visualization were used for better decision-making regarding high-stress areas.
- The proposed method enhances the traditional digital twin model by considering human perception as an important factor for planning.

FUTURE WORK:

- Use of real-time data integration technology like IoT sensors to create dynamic digital twin models.
- Expansion of the system to accommodate large-scale simulations for cities.
- Use of advanced machine learning algorithms to enhance the accuracy of the results from the perception predictions.
- Use of GIS technology to enhance the accuracy of the results.
- Use of user feedback data to validate the results from the perception predictions.
- Development of multi-objective optimization models.

9. ADVANTAGES OF PROJECT

- **Human-Centric Planning:** Human perception of stress and comfort is included in infrastructure development planning, allowing for better urban infrastructure that is people-oriented.
- **Decision Support System for Data:** Various environmental variables are combined to quantify urban infrastructure status and interventions.
- **Evaluation Through Scenario Simulation:** The tool allows planners to create virtual scenarios of various urban infrastructure changes and test their effectiveness before actual urban development.
- **Visual Spatial Representation:** The system creates a visual representation of the urban environment through heat maps and dashboards.
- **Risk Mitigation and Cost Effectiveness:** It allows planners to test various solutions without the risk of costly and wasteful physical infrastructure modification.
- **Flexibility and Scalability:** The tool can be used in any urban environment and adapted to any data set.
- **Primary Beneficiaries:** Urban planners, local governments, policymakers, researchers, and the citizens themselves would greatly benefit from the tool.

10. OUTCOME

The present project has led to an outcome that is focused on research, wherein the perception-based digital twin architecture for designing human-centric urban infrastructure planning has been developed and described in detail. An attempt has been made to submit the research to Bentham Science in order to get it published in a peer-reviewed journal.

It is possible to state that the present research adds value to the domain of smart cities due to the inclusion of perception in a digital twin framework. Such an outcome indicates the possibility of extending the project in the directions of academic contributions and research.

The screenshot displays the 'MPS' (Powered by Eureka) Editorial Board Member dashboard. The interface includes a sidebar with navigation links: User Portal, Dashboard, New Submission, Email Notifications, Support, and User Manuals. The main content area is divided into three sections: 'ASSOCIATE AUTHOR SUBMISSIONS', 'NEWLY SUBMITTED ARTICLES', and 'AWAITING REVISION'. The 'ASSOCIATE AUTHOR SUBMISSIONS' section shows a table with columns: Journal, Reference No., Date, Title, and Status. It indicates 'No Submission' and shows 'Showing 0 to 0 of 0 entries'. The 'NEWLY SUBMITTED ARTICLES' section shows a table with columns: Journal, Reference No., Date, Title, and Status. It lists one article: RACSC, BMS-RACSC-2026-193, dated 2026-04-03, titled 'An Emotion-Aware Digital Twin for Human-Centric Urban Infrastructure Planning', with a status of 'Awaiting Editorial Approval'. It shows 'Showing 1 to 1 of 1 entries'. The 'AWAITING REVISION' section shows a table with columns: Journal, Reference No., Date, Title, Status, and Action. It indicates 'No Submission' and shows 'Showing 0 to 0 of 0 entries'.

Journal	Reference No.	Date	Title	Status
RACSC	BMS-RACSC-2026-193	2026-04-03	An Emotion-Aware Digital Twin for Human-Centric Urban Infrastructure Planning	Awaiting Editorial Approval

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